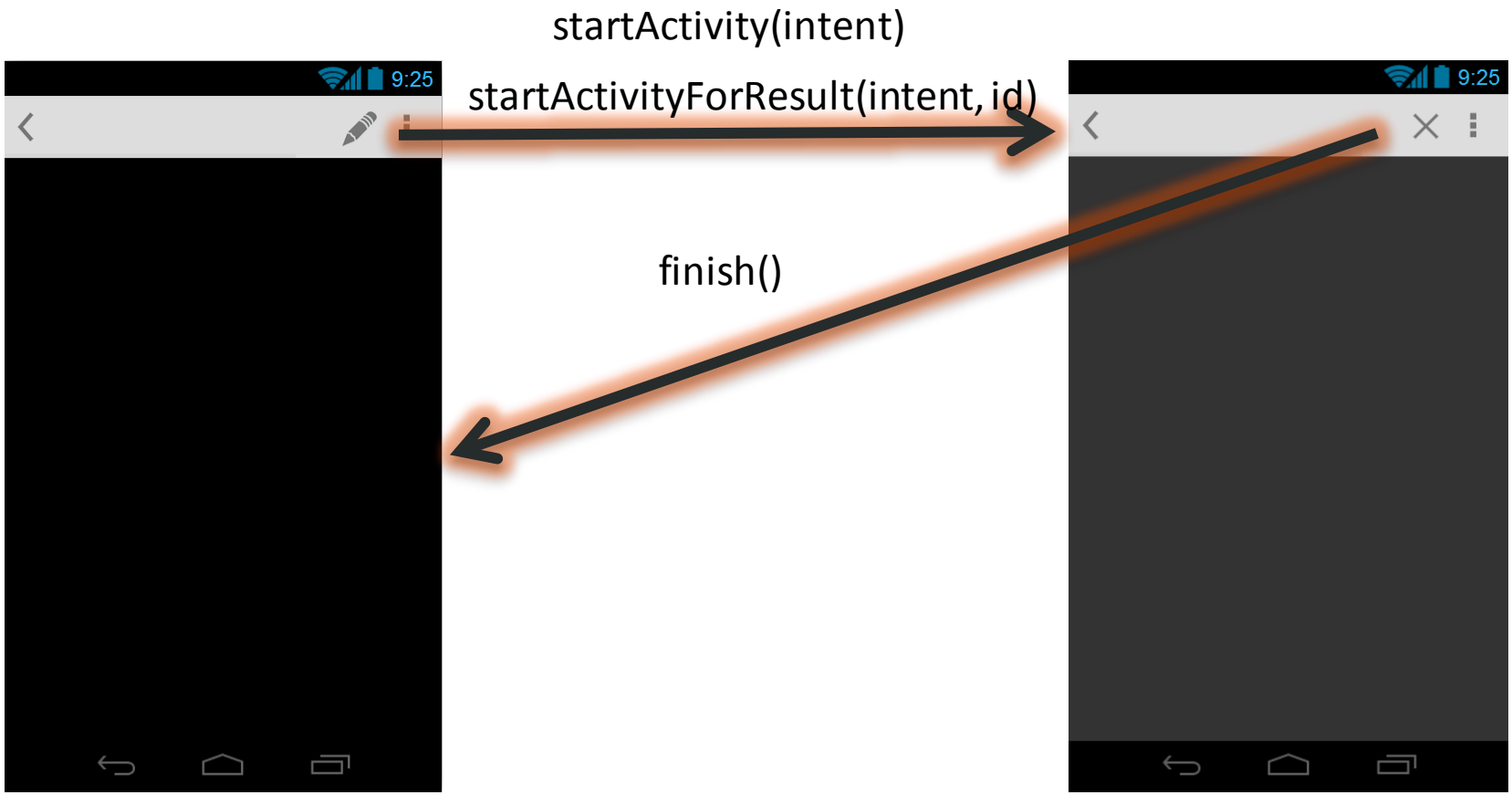
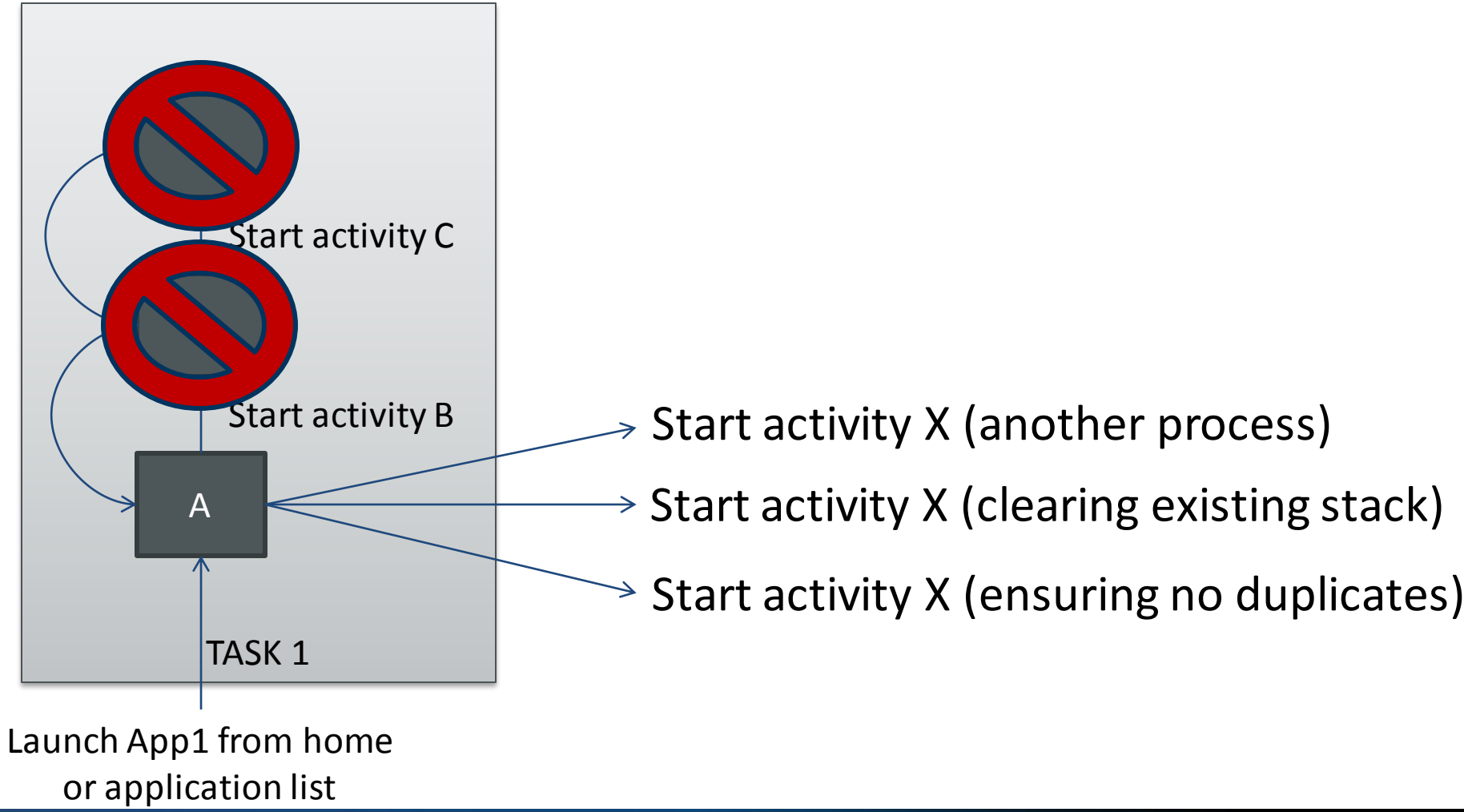
605.486 – Android Development

Module 3a
Intents

Activity Switching



Back Stack

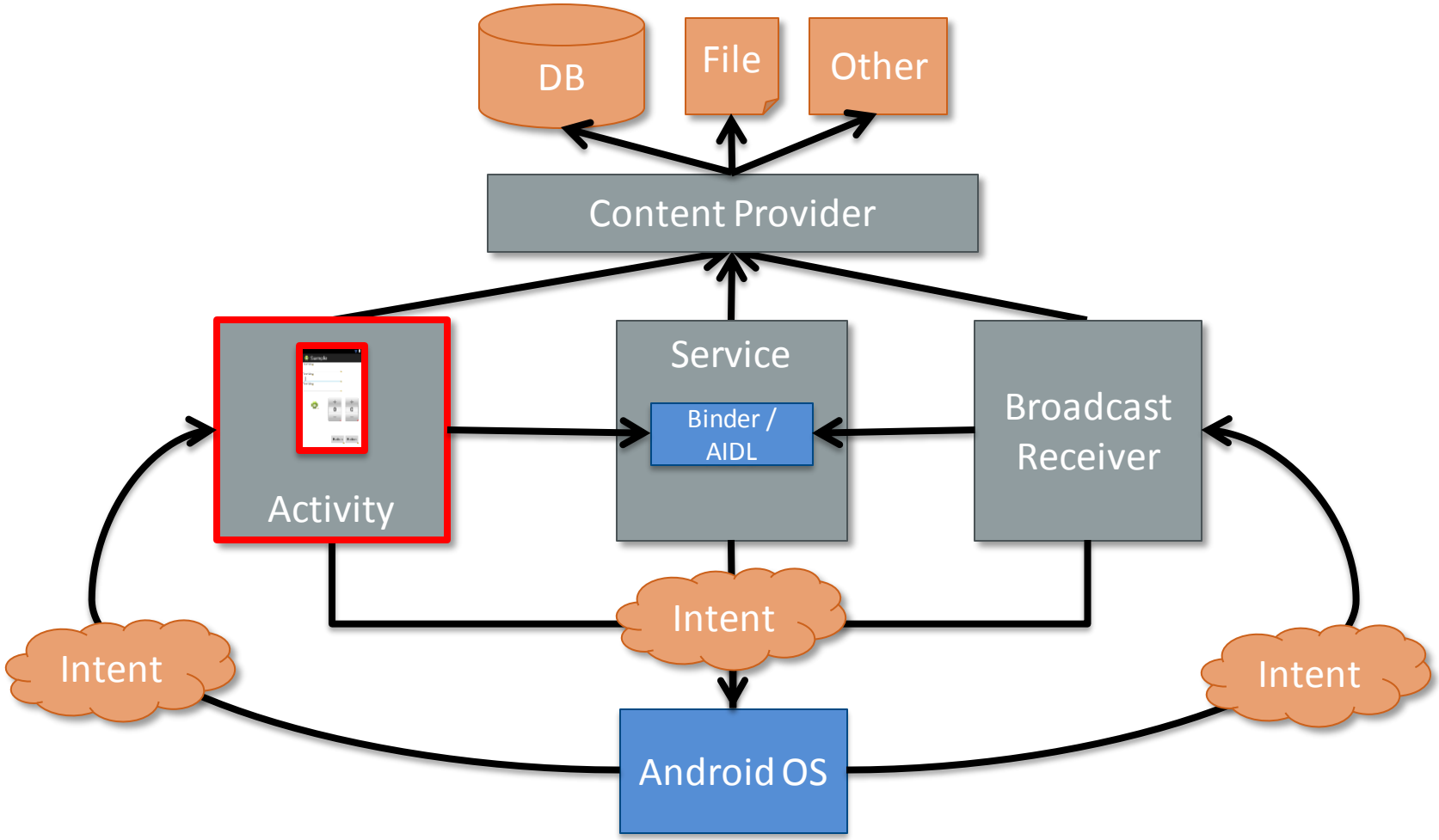

Start activity C
Start activity B
A
TASK 1
Launch App1 from home
or application list
Start activity X (another process)
Start activity X (clearing existing stack)
Start activity X (ensuring no duplicates)

Going back…

- DO NOT CALL startActivity() TO GO BACK!
  - ***<u>You don't know where you came from!</u>***
- Target pushed on the back stack!
  - When going back, you want to pop!
  - Use finish()
  - Remember that this won't actually go back until the current method ends!

Application Components

and Typical Communication



Intent Types

- Explicit – I want to start a specific activity
- Implicit – describes what to do
  - Action – view, edit, call, scan, play…
  - Category – preference, launcher, gadget, browsable…
  - Mime Type – image, html, text, mp3…
  - Url – http://javadude.com

Explicit Intent

```
Intent intent =
    new Intent(this, SomeActivity.class);

startActivity(intent);
```

- You do not need an Intent Filter to receive this

Implicit Intent

```
Intent intent =
   new Intent(Intent.ACTION_EDIT);
intent.setType("application/vnd.javadude.data");
intent.putExtra("id", 42);
startActivity(intent);
```

- Category DEFAULT added automatically implicit intents!
  - Any intent filters that want to catch implicit intents must include DEFAULT as a category
  - exception: MAIN/LAUNCHER intents do not have DEFAULT

Implicit Intent Matching

Intents are matched to Intent Filters

- Action: Intent action must match one action in filter
  - Filter may have additional actions
- Category: All intent categories must be present in filter
  - Filter may have additional categories
- Data (URI pattern/mime type)

Implicit Intent Matching

Intent has URI	Intent has type	Filter has URI	Filter has type	Pass?
no	no	none	none	Yes
yes	no	yes	no	yes
no	yes	no	yes	yes
yes	yes	yes, or intent is **content:** or **file:**	yes	yes

MIME Intent Filter

```
<!-- MIME filter -->
<intent-filter>
    <action android:name="android.intent.action.EDIT"/>
    <category
      android:name="android.intent.category.DEFAULT"/>
    <data
      android:mimeType="application/vnd.javadude.data"/>
</intent-filter>
```

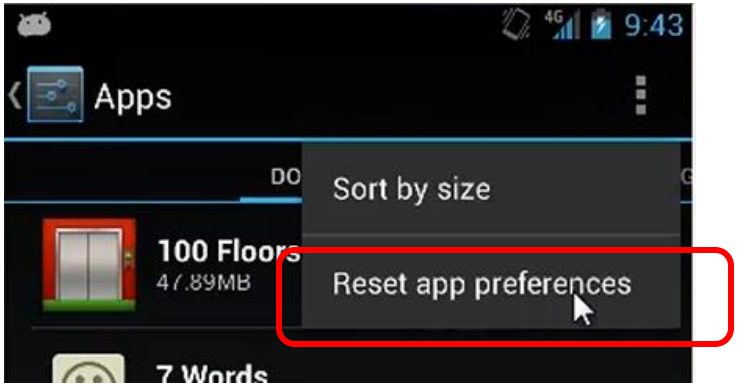
URI Intent Filter

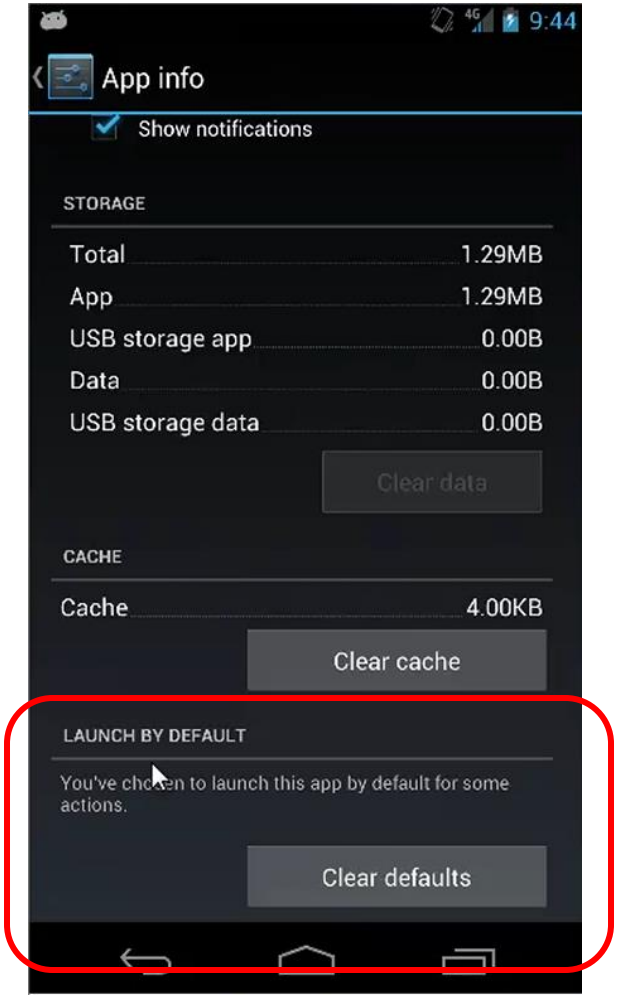
```
<!-- URI filter -->
<intent-filter>
    <action android:name="android.intent.action.EDIT"/>
    <category
      android:name="android.intent.category.DEFAULT"/>
    <data android:scheme="http"
          android:host="www.javadude.com"/>
</intent-filter>
```

MIME and URI Intent Filter

```
<!-- Both -->
<intent-filter>
    <action android:name="android.intent.action.EDIT"/>
    <category
        android:name="android.intent.category.DEFAULT"/>
    <data android:mimeType="application/vnd.javadude.data"
        android:scheme="http"
        android:host="www.javadude.com"/>
</intent-filter>
```

When Multiple Filters Match





Passing Data

- Can attach “extras” to Intents
  - Primitives / Serializable / Parcelable (later)
  - *Data is marshaled; target activity could be in different process*
- Target Activity can access
  - getIntent().getIntExtra(“foo”, -1)

Returning Data

- Caller
  - calls startActivityForResult(intent, requestId)
  - overrides onActivityResult(request, result, data)
- Target calls
  - setResult(RESULT_OK, intent)
  - setResult(RESULT_CANCELED, intent)